XRPL Validator — Daily Engineering Report · 2026-09-12

Period: 2026-09-12 00:00 to 24:00 US Eastern (2026-09-12 04:00 to 2026-09-13 04:00 UTC); ledgers #106,926,932 to #106,949,449. Every count from the ledger and the validator's logs stops at #106,949,449. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 5 engine receipts recorded (4 fixed, 0 explained, 1 without a recorded fix).
- **Changes:** 0 deployments, 2 commits merged into main, 1 engine finding fixed (F266).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Ledgers with a receipt in the period (#106,926,932–#106,949,449), by step:

Step	Ledgers with a receipt
Step 1 · native engine	5
Step 2 · records	0
Step 3 · order	0

Step	Ledgers with a receipt
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#106,930,301	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-02	—	#106,937,018	Step 1 · native engine	1 transaction result(s) differed from rippled's; 8 ledger change(s) the network made were absent from ours: 8EE34BE9, B6E0EC5A, 19D07933; 1 object(s) differed in content	fixed: F266, commit 357975b, deployed in cycle 124 (2026-09-13 05:42 UTC)
R-03	—	#106,945,777	Step 1 · native engine	1 transaction result(s) differed from rippled's; 10 change(s) of ours the network did not make; 2 object(s) differed in content	fixed: F267, commit e094350, deployed in cycle 124 (2026-09-13 05:42 UTC)
R-04	—	#106,945,810	Step 1 · native engine	1 transaction result(s) differed from rippled's; 11 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F267, commit e094350, deployed in cycle 124 (2026-09-13 05:42 UTC)
R-05	—	#106,948,593	Step 1 · native engine	6 object(s) differed in content	fixed: F268, commit 123e9a7, deployed in cycle 126 (2026-09-13 15:07 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-12-01, INC-01 is INC-2026-09-12-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployments.

Merged into main

Merged	Repo	Commit	Subject
15:40 UTC	xrpl-validator	`b0ec4bd`	chore(watch_engine): ter-miss breakdown counts this run's receipts; upstream panel; no LAN IP
03:13 UTC	xrpl-validator	`357975b`	fix(xrpl-ledger): F266 — a token escrow finished by its destination creates the missing trust line

Findings fixed

- **fix(xrpl-ledger): F266 — a token escrow finished by its destination creates the missing trust line** (`357975b` , xrpl-validator) rippled's `escrowUnlockApplyHelper<Issue>` (EscrowHelpers.h) runs `trustCreate` when the receiver has no line and `createAsset` — which EscrowFinish sets to `destID == accountID_` (EscrowFinish.cpp:368): the destination finishing its own escrow accepts the line. The line lands with limit zero, NoRipple per the destination's DefaultRipple, the reserve on its side, both owner directories and OwnerCount + 1, after …

Docs and operations

- **chore(watch_engine): ter-miss breakdown counts this run's receipts; upstream panel; no LAN IP** (`b0ec4bd` , xrpl-validator)

Independence program

No independence record for UTC day 2026-09-12; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

No amendment held a majority.

Amendments enabled on mainnet: 94.

Open items

- Receipt R-01 (#106,930,301): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Size-Weighted Owner Reserve for Big Objects** (fees) Add a computed SField sfReserveUnits (UInt8, capped at 8) to any variable-length ledger object — starting with MPTokenIssuance, whose XLS-89 metadata
- **Fixed-Term Amortizing Loans with Rate Oracle** (lending) Add sfMaturityCloseTime, sfDayCountBasis, and an optional sfRateOracleID (a PriceOracle keylet) to the Loan object introduced by LendingProtocol, plus
- **Segregated Sub-Accounts with MPC Quorum Custody** (custody) Add a new owner-reserved ltSUB_ACCOUNT object (keylet::subAccount(parent, seq)) holding its own XRP balance and trust-line/MPT directory, created by S

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-12 15:40	09-12 15:40	xrpl-validator	`b0ec4bd`	chore(watch_engine): ter-miss breakdown counts this run's receipts; upstream panel; no LAN IP
09-13 03:13	09-13 03:13	xrpl-validator	`357975b`	fix(xrpl-ledger): F266 — a token escrow finished by its destination creates the missing trust line

Times in UTC. A commit appears on the day it was committed to main or merged into it.